

ST MARY 1 SSW, MT, USA

WMO: 742290

Lat: 48.7412N

Lon: 113.4330W

Elev: 4555

StdP: 12.43

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 04130

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.1	-11.7	-24.9	1.6	-18.0	-18.9	2.3	-12.3	21.7	33.0	19.9	31.8	2.2	N/A	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.6	82.5	56.3	79.6	55.6	76.6	55.0	59.5	74.7	57.9	74.4	56.2	73.1	6.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.6	72.4	62.9	51.3	66.4	62.4	49.2	61.5	61.6	28.1	74.7	26.9	74.0	25.7	73.4	66.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	17.4	15.6	-28.1	88.3	5.1	3.9	-31.8	91.1	-34.7	93.4	-37.6	95.6	-41.3	98.5
			-29.6	62.7	4.6	2.3	-32.9	64.4	-35.6	65.7	-38.2	67.0	-41.5	68.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.9	24.1	22.0	30.0	37.0	45.6	52.4	60.7	58.8	50.8	41.2	31.3	23.2	(6)
(7)		DBStd	16.76	14.36	14.26	11.19	8.24	7.18	5.80	5.06	5.64	7.78	9.49	12.42	13.84	(7)
(8)		HDD50	4600	804	785	621	394	175	39	1	7	83	296	563	832	(8)
(9)		HDD65	9199	1268	1205	1086	841	601	378	149	199	426	739	1010	1297	(9)
(10)		CDD50	902	0	0	0	3	39	112	334	281	108	22	3	0	(10)
(11)		CDD65	27	0	0	0	0	0	1	17	8	1	0	0	0	(11)
(12)		CDH74	1207	0	0	0	0	7	69	603	434	92	2	0	0	(12)
(13)		CDH80	228	0	0	0	0	0	9	130	81	8	0	0	0	(13)
(14)	Wind	WSAvg	7.0	9.3	7.5	7.7	6.8	5.8	6.1	5.3	5.0	5.8	7.5	9.0	8.7	(14)
(15)	Precipitation	PrecAvg	22.4	1.5	1.3	1.5	1.8	2.8	3.8	1.7	1.5	1.7	1.6	1.9	1.5	(15)
(16)		PrecMax	32.8	3.5	2.8	3.3	3.3	5.6	7.9	6.5	3.4	3.7	3.9	5.7	4.1	(16)
(17)		PrecMin	14.2	0.2	0.2	0.6	0.4	0.6	1.2	0.1	0.0	0.1	0.1	0.1	0.3	(17)
(18)		PrecStd	5.0	0.8	0.7	0.8	0.7	1.5	2.2	1.6	1.0	1.0	1.0	1.5	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	49.9	55.5	65.3	74.6	80.8	88.2	86.4	81.1	70.8	59.8	50.8	(19)
(20)			MCWB	37.2	38.4	41.0	45.3	51.4	56.3	59.1	56.7	54.5	49.9	44.3	38.4	(20)
(21)		2%	DB	47.7	46.4	50.4	59.5	69.6	75.2	83.6	82.1	76.7	63.3	54.8	46.0	(21)
(22)			MCWB	36.7	37.1	38.7	43.3	50.1	56.5	56.5	55.9	53.4	47.4	42.0	35.5	(22)
(23)		5%	DB	43.6	43.2	47.1	55.0	66.0	71.0	80.4	79.1	72.7	59.4	50.1	42.8	(23)
(24)			MCWB	34.8	35.7	37.6	41.7	48.9	53.6	56.5	55.0	52.1	45.0	40.7	34.8	(24)
(25)		10%	DB	40.5	39.6	44.3	50.9	61.4	66.9	77.3	75.5	67.8	56.0	46.2	39.3	(25)
(26)			MCWB	33.1	33.5	36.2	39.7	46.9	51.6	55.7	54.5	50.5	43.5	38.0	33.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	39.5	42.2	47.8	53.1	60.3	63.7	60.3	57.9	51.4	45.9	40.4	(27)
(28)			MCDWB	49.4	47.7	51.5	63.9	70.7	71.3	76.8	74.1	70.9	64.0	57.1	48.4	(28)
(29)		2%	WB	37.1	37.9	40.1	44.4	51.1	57.1	60.2	58.6	55.2	48.6	44.3	37.3	(29)
(30)			MCDWB	44.5	44.9	48.1	58.0	66.9	71.9	75.5	75.4	71.1	60.7	54.7	42.7	(30)
(31)		5%	WB	35.1	36.1	38.1	41.9	49.6	54.3	58.6	57.0	53.3	46.2	41.4	35.0	(31)
(32)			MCDWB	41.4	43.0	45.8	53.0	64.4	68.1	74.7	75.1	70.2	56.8	49.7	41.1	(32)
(33)		10%	WB	33.4	33.0	36.2	39.8	47.8	52.4	57.0	55.5	51.3	44.1	38.4	33.0	(33)
(34)			MCDWB	39.4	38.8	43.3	49.1	61.7	64.0	73.6	72.9	67.3	54.7	45.3	38.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.0	16.3	15.4	16.3	19.3	19.9	26.6	26.8	21.8	15.5	12.8	13.1	(35)
(36)			MCDBR	16.0	16.9	19.2	26.0	31.0	31.5	35.7	37.2	35.5	23.8	15.9	13.8	(36)
(37)			MCWBR	11.0	7.9	11.9	14.3	15.3	14.9	15.7	15.5	15.8	12.4	9.0	8.1	(37)
(38)		5% WB	MCDWB	15.9	14.2	16.6	23.6	27.9	28.3	31.9	32.1	30.8	21.0	15.1	12.0	(38)
(39)			MCWBR	12.0	9.9	10.4	13.1	13.8	14.6	15.3	14.5	14.7	11.8	9.3	8.1	(39)
(40)		Clear-Sky Solar Irradiance	taub	0.231	0.238	0.259	0.294	0.304	0.308	0.308	0.318	0.295	0.276	0.246	0.228	(40)
(41)	taud		2.401	2.402	2.432	2.405	2.505	2.547	2.560	2.514	2.554	2.578	2.526	2.406	(41)	
(42)	Ebn at Noon		276	299	308	304	304	302	300	293	292	282	269	262	(42)	
(43)	Edh at Noon		20	25	30	34	32	31	30	30	26	22	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	392	708	1087	1492	1778	1992	2264	1829	1300	791	432	314	(44)	
(45)		RadStd	31	53	102	94	161	164	138	123	107	74	31	26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page